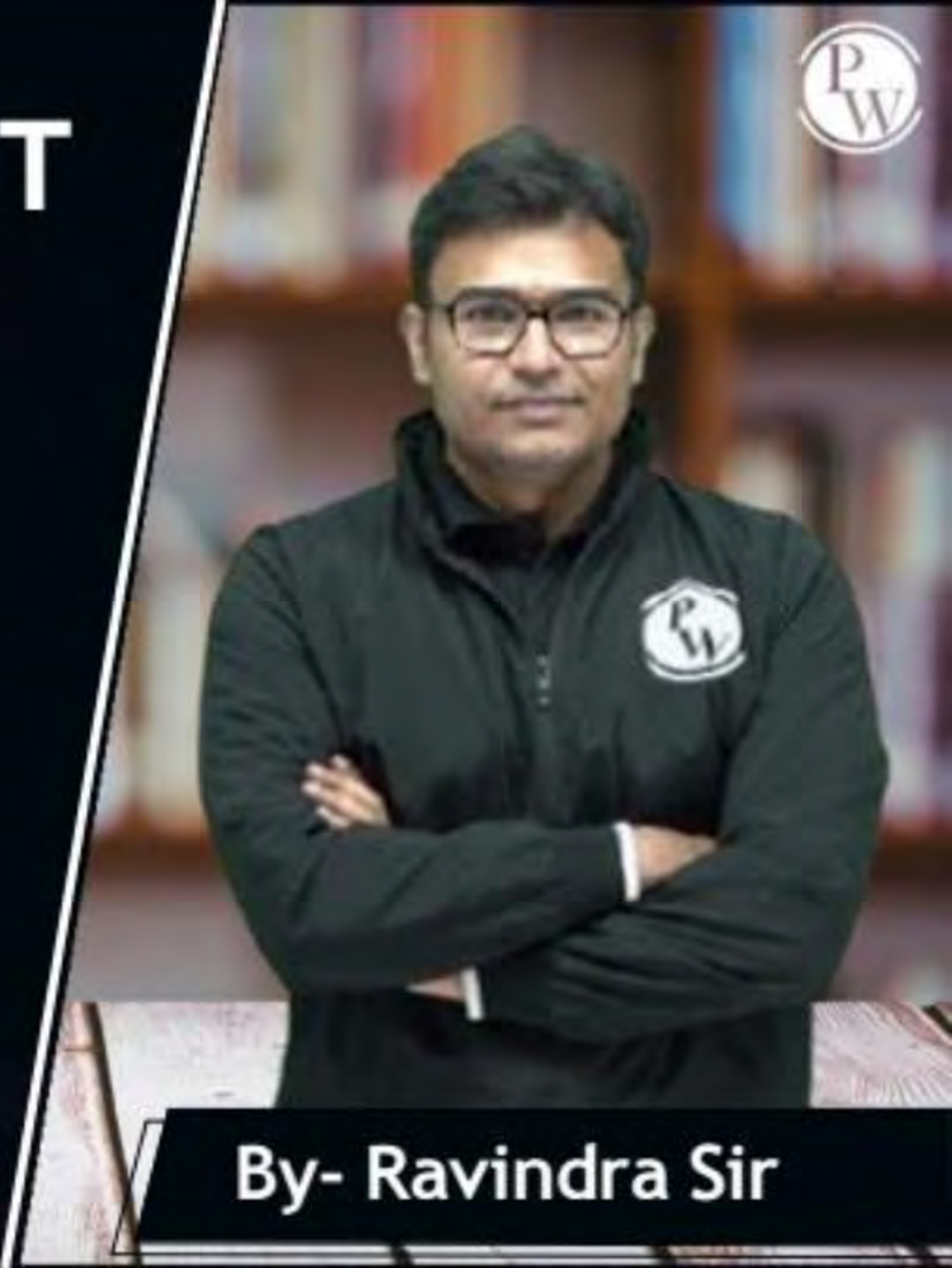
Computer Science & IT

COMPUTER NETWORK (CN)

TCP and UDP

Lecture No. 03



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Basic algo

$TO = (2) * RTT$

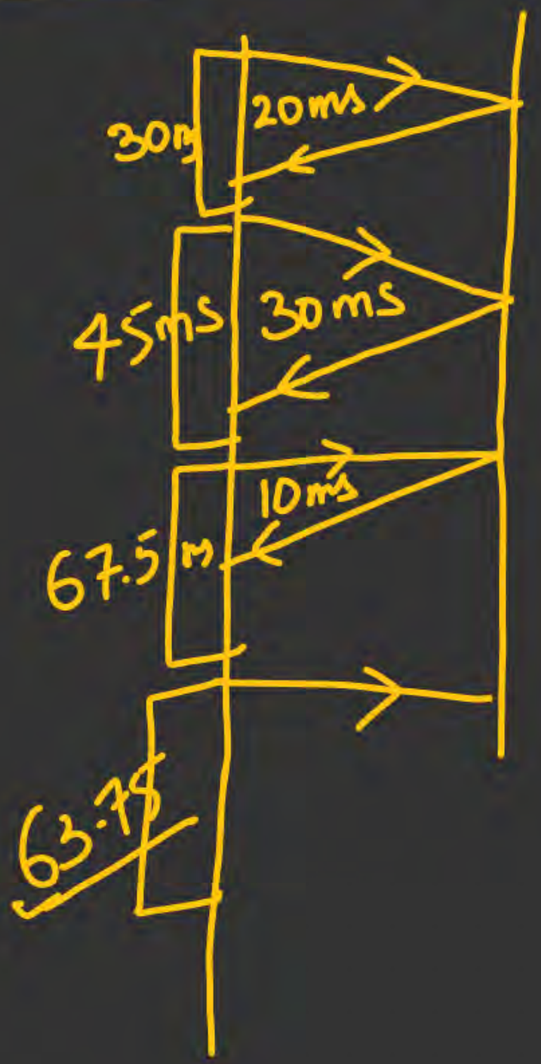
Topics to be Covered



- Topic
- Topic
- Topic

Jacobsons algo

Jacobson's algo :-



IRTT = 10ms ✓ 10ms ± 5

ID = 5 ms 5ms-15m

TO = 4 × D + RTT

= 4 × 5 + 10 = 30 ms

ARTT = 20ms

AD = |10 − 20| ms = 10 ms

✓ NRTT = α(IRTT) + (1 − α) ARTT

(α = 0.5)

= 15ms ✓

✓ ND = α(ID) + (1 − α) AD

= 7.5 ✓

(2) IRTT = 15 ms ✓

ID = 7.5 ms

TO = 4 * D + RTT

= 45 ms

ARTT = 30 ms ✓

AD = |IRTT - ARTT| = |15 - 30| = 15 ms

✓ NRTT = (0.5)(15 ms) + (0.5)(30) ms

= 22.5 ms

✓ ND = (0.5) 7.5 + (0.5)(15) = 11.25 ms

(3) IRTT = 22.5 ms

ID = 11.25 ms

TO = 4 * D + RTT

= 67.5 ms

ARTT = 10 ms

AD = 12.5 ms

✓ NRTT = α(IRTT) + (1 - α)(ARTT)

= 16.25 ms

✓ ND = α(ID) + (1 - α)AD)

= 11.875

④ IRTT = 16.25

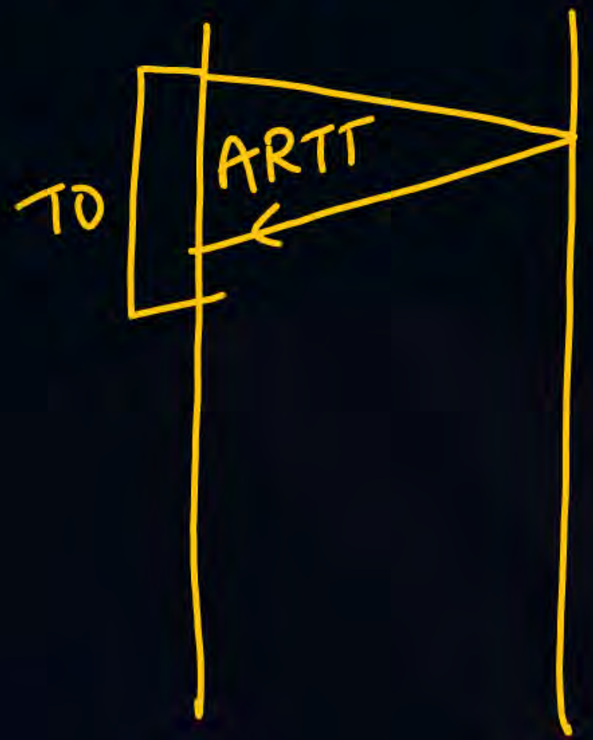
ID = 11.875

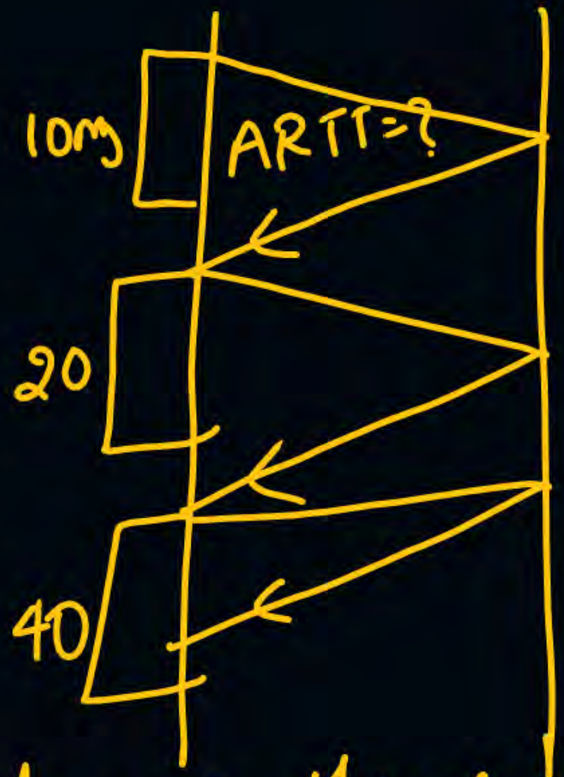
TO = 4 × D + RTT

= 63.75

Gate:- IRTT = 10 ms, ID = 5 ms, $\alpha = 0.5$, Successive ARTT are 20, 30, 10 ms, then what is the TO for 4th packet

Disadv of both basic and Jacobson's algo

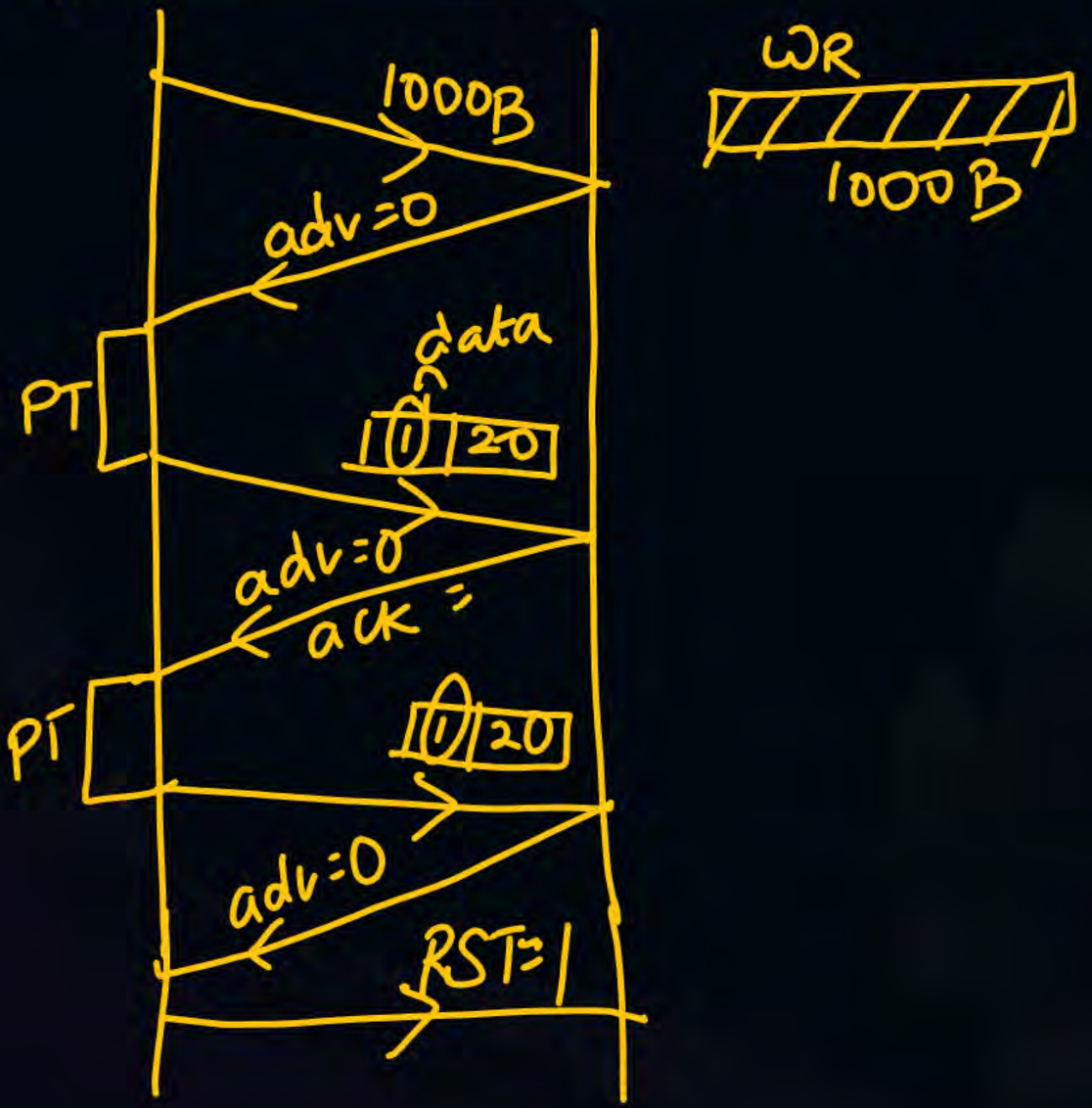




Karn's modification, whenever there is a TO, simply double the TO times

Silly window Syndrome :-

① when receiver Sends WIN=0



② when Sender has only Small amount of data to send



Nagle Solution:-

Wait for 1RTT and whatever data we can accumulate, Send that data

In case we get 1mss before 1RTT then send 1mss immediately

In case we get less than 1mss, then send it.

Gate Ex:- In nagles algo, Sender is sending 1Byte/ms RTT=100ms, mss=200B, then how much data will the Sender send. $W_S = 1RTT \times 1B/ms$
$= 100 \times 1 = \underline{100B}$.

③ Destination is Consuming only 1 Byte at a time





Clark's Solution:-

Receiver should wait until 1mss or ½ buff is free, then send adv window

Traffic shaping :-

1) Leaky bucket
2) Token bucket

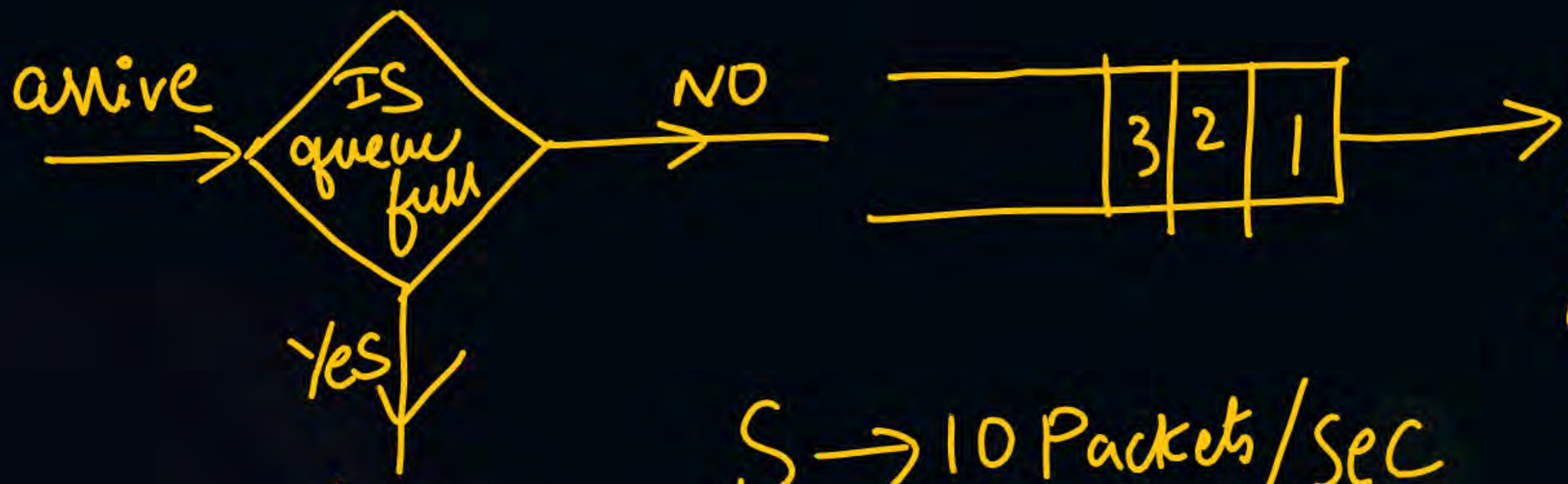
This is another method of congestion control to shape the traffic before it enters the N/W.

Traffic shaping controls the rate at which packets are sent

→ During the connection establishment, the sender and the carrier will negotiate traffic pattern

Leaky bucket :-

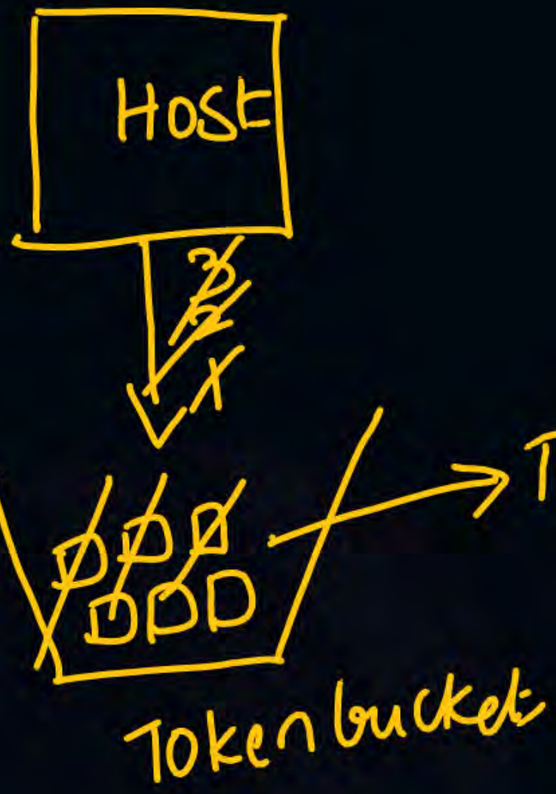




S → 10 Packets / sec

MESS




Host
Tokens get accumulated.
Token bucket

Gate:- let the initially capacity of token bucket is 'C' tokens and tokens enter at a rate of 'γ' tokens/sec. The maximum number of packets that enter the N/w during First t sec

$C + \gamma t$

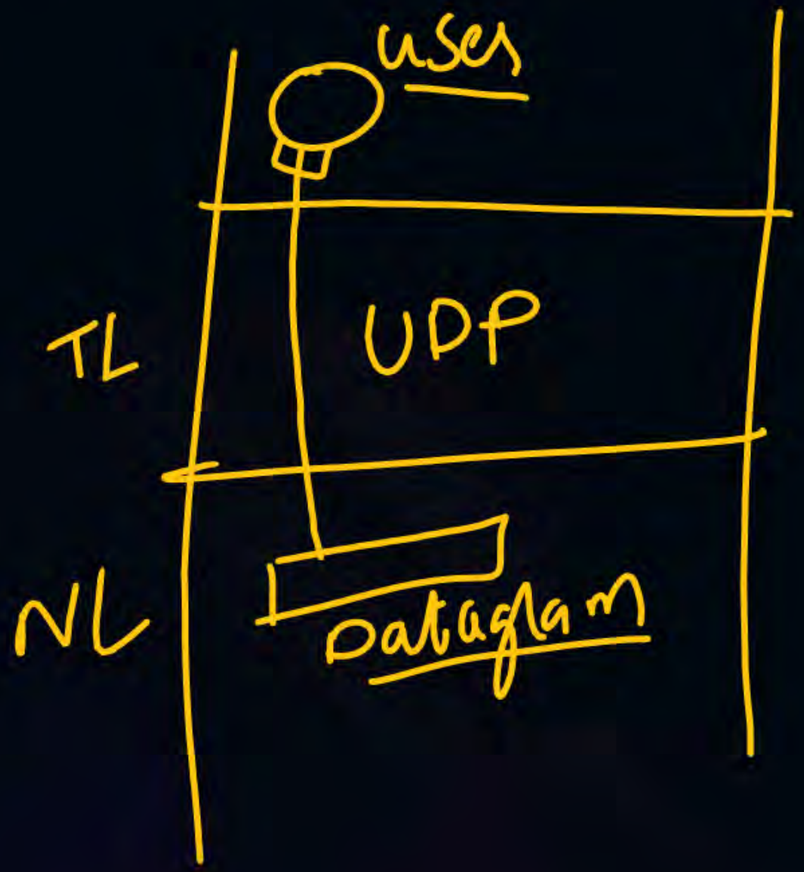
Packets



t sec

$C + \gamma t$ → packets

UDP :- User datagram protocol



Need of UDP :-

1) Applications that need 1 req 1 rep ✓



✓



1) DNS ✓
2) BOOTP
3) DHCP
4) NTP X
5) NNP X
6) quote of the day X

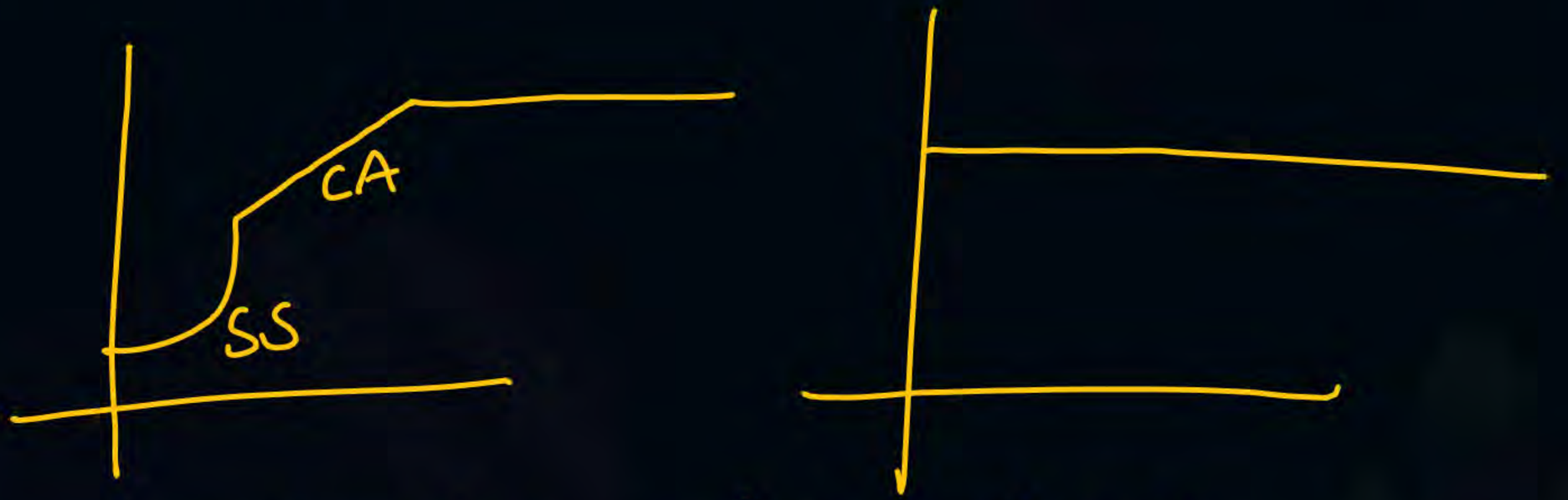
2) Broadcast and multicast ✗



2^{24} → TCP → 2^{24} Connections

UDP ✓

3) Some applications need speed than reliability



multimedia, online gaming ✓

UDP :-



header

00000

Length of UDP header + UDP data

UDPH + UDPD + Pseudo header from IP

many applications dont use checksum

DHCP, DNS

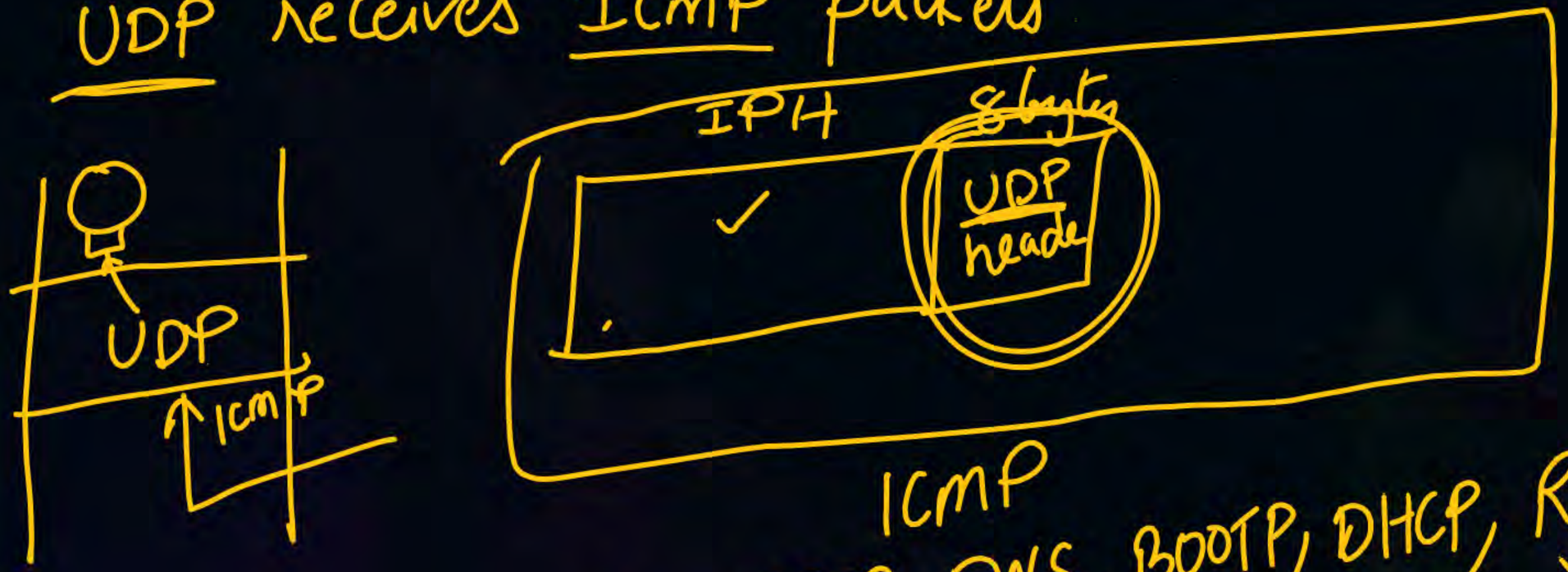
Seq^{X}, ack^{X}, SYNX, FINX, RSTX, URGX, PSHX

options X

UDP will have facility to set options in IP.

1) Traceroute 2) Route route

UDP receives ICMP packets



ICMP

Ex: TFTP, DNS, BOOTP, DHCP, RIP, OSPF
X X

Hardware devices in CN:-

Cables :-

- 10 Mbps
  - 10 Base T → 100 m
  - 10 Base 2 → 200 m
  - 10 Base 5 → 500 m
- 100 Mbps
  - 100 Base T → 100 m

Collion domain = n



✓ Base band → only one signal at time
→ collisions are possible

(PL) ✓ (Attenuation)

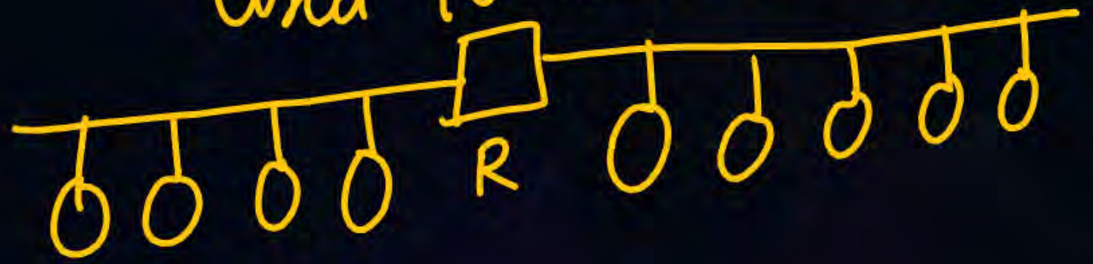


Repeaters and hubs :- → PL



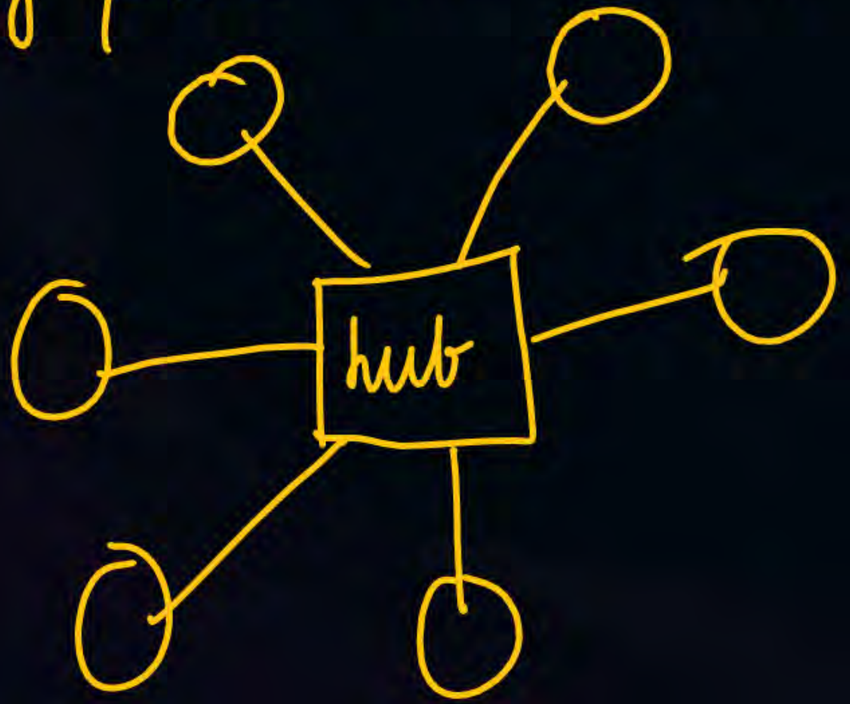
A repeater is a device that operates at PL
Signals that carry information within a n/w can travel
a fixed distance before attenuation endangers the
integrity of the data

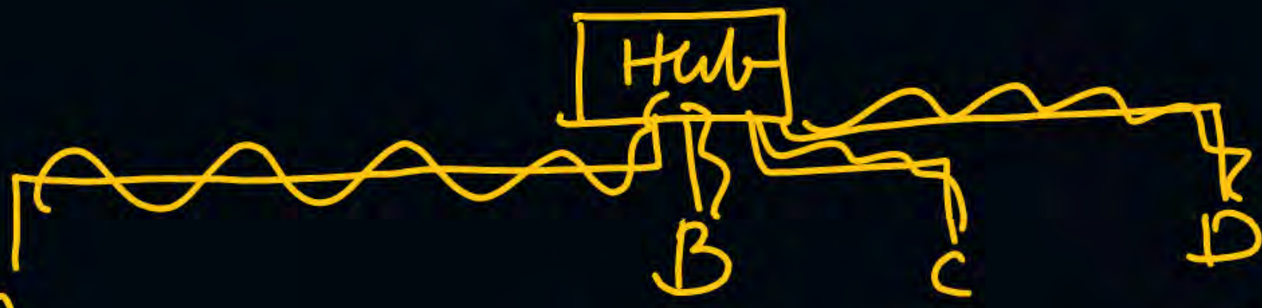
A repeater receives signal and before it becomes too weak or corrupted, regenerates and retransmits the signal. The repeater sends refreshed signal. In the past Ethernet LANs were using bus topology, a repeater was used to connect two segments of the LAN, to overcome the length restrictions of the cable



Today, however ethernet LANs use Star topology

In Star topology, a repeater is a multiport device often called as hub, that can be used to serve as a connecting point and at the same time function as as a repeater





when a packet is sent from Station A to Stat B, the signal representing the frame is regenerated removing noise and forwards it to all outgoing ports

In other words, the frame is broadcasted.

All stations receive the frame but only B keeps it

The rest of the stations discards it.

Hub does not have DLL, so it cannot see MAC addresses

So it sends the packet on all ports



Collision domain 'N'



Collisions can happen inside a hub

collision domain → 'n'

Hub :-

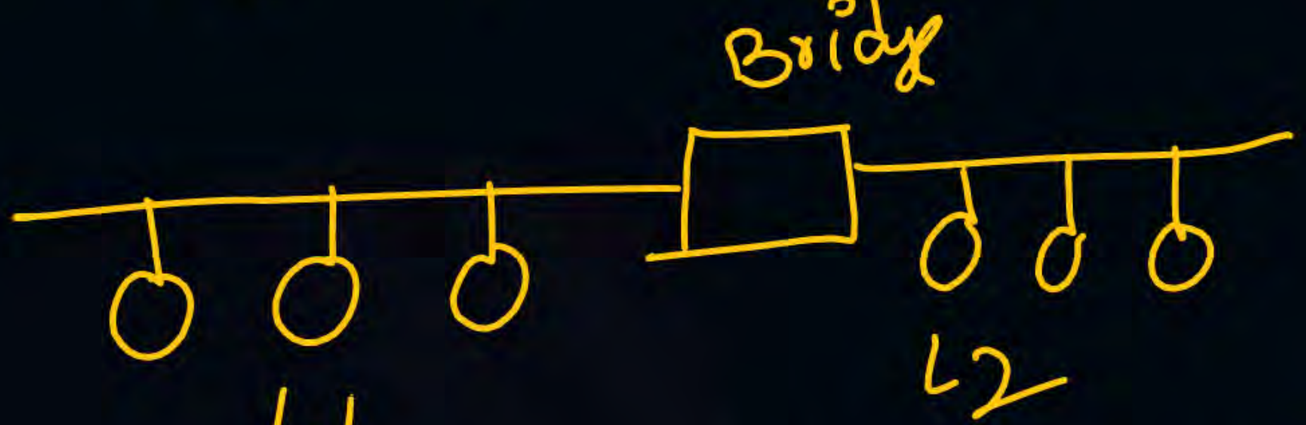
Dis :- 1) Collisions ✓

2) Heavy traffic ✓

Adv :- low cost ✓

Bridges :- (Costly)

A single Organisation may end up in multiple LANs



All these lans need interaction, so Bridges are needed

Seperate LANS can be connected to one Bridge making it one big LAN

Ex:- many ethernet LANs can be connected to one bridge



The main difference b/w hub and bridge is that bridge has DLL, can see MAC addresses, so avoids broadcasting

Ideally Bridges should be transparent.

It should be possible to go out and buy Bridges, plug the LAN cables into the Bridge and have everything working perfectly instantly. There should not be any hardware changes, software changes, no Setting of addresses, no downloading of Routing table & parameters, nothing all.

Just plug the cables and walk away.

Further more, the operation of existing LANs should not be affected by bridges at all. Even stations should not observe any difference. It should be easy to to move stations around the bridged LAN's, as if it is to move them around a single LAN.

THANK - YOU